

Spec-FastGS: Enhance Specular Reconstruction Performance in Gaussian Splatting using Guided-optimization Anisotropic Spherical Gaussians

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Abstract—3D Gaussian Splatting (3DGS) has emerged as a groundbreaking technique for real-time rendering of novel views, offering exceptional visual quality and speed. However, the standard 3DGS method struggles to accurately reconstruct specular surfaces, leading to artifacts and loss of detail in reflective areas. While recent variants address this by integrating complex directional functions, they suffer from severe computational overhead and excessive VRAM consumption. In this paper, we propose Spec-FastGS, a highly efficient 3DGS architecture that accurately reconstructs specular highlights without sacrificing rendering speed or memory. Our method introduces a hybrid appearance representation that seamlessly separates diffuse and specular components, leveraging Spherical Harmonics (SH) alongside Anisotropic Spherical Gaussians (ASG). To optimize computational resources, we design a prior-guided densification mechanism that intelligently allocates ASG capacity only to highly reflective regions. Extensive experiments on complex datasets demonstrate that Spec-FastGS achieves state-of-the-art qualitative and quantitative improvements in specular reconstruction while maintaining the fast training and real-time rendering capabilities of its underlying architecture.

I. INTRODUCTION

Reconstructing high-fidelity digital replicas of real-world environments is a foundational challenge in computer vision and 3D graphics, serving as the backbone for modern applications ranging from immersive virtual reality to industrial digital twins. Innovations such as Neural Radiance Fields (NeRF) have achieved substantial breakthroughs in generating novel views of 3D scenes, presenting the potential for photorealistic renderings. Despite these advancements, NeRF-related methodologies still contend with challenges such as computationally expensive optimization and slow rendering speeds. These limitations restrict their application in real-time interactive scenarios.

More recently, 3D Gaussian Splatting combines an explicit point-based 3D representation with tile-based splatting techniques to achieve high-quality 3D scene modeling and real-time rendering. However, it suffers from a severe performance drop on scenes featuring specular and reflective surfaces. This

is because standard 3D Gaussian Splatting relies on Spherical Harmonics (SH), a low-frequency representation that fails to capture the sharp, view-dependent changes characteristic of specular highlights found on metals, glass, or glossy furniture.

Accurately modeling these reflective surfaces is a challenging task. Recent variants, such as Specular-Gaussians, attempt to overcome this by replacing SH with complex optical functions like Anisotropic Spherical Gaussians (ASG). However, they suffer from time-consuming optimization and slow rendering speeds. Because they naively apply heavy ASG computations globally across all primitives, they experience exploding VRAM consumption. It is still an unexplored problem how to combine complex anisotropic shading in a 3D Gaussian framework to improve reflections while preserving the extreme efficiency in training and rendering.

In this paper, we present Spec-FastGS, a novel method that enhances the neural rendering of 3D Gaussians on reflective surfaces while preserving the low VRAM footprint and real-time rendering speeds of its underlying architecture. Built upon the memory-efficient FastGS backbone, our method introduces a Hybrid Appearance Representation that decouples the scene’s lighting. We retain SH for modeling smooth diffuse regions while specifically targeting specular highlights with ASG, allowing the model to simulate a wide array of real-world intricate lighting situations.

The main challenge in applying this hybrid representation lies in preventing computational bloat and optimization instability. In Spec-FastGS, we address this problem by introducing a novel Reflection-Prior Guided Densification mechanism. By extracting a 2D reflection probability mask (Ref_Score), our pipeline intelligently restricts the allocation of heavy ASG parameters exclusively to regions with true reflective properties. Furthermore, we implement a Role Separation Supervision strategy—utilizing gradient masking and dedicated specular loss—to prevent interference between the diffuse and specular components during optimization.

Built upon FastGS, Spec-FastGS maintains real-time ren-

dering speeds while accommodating complex view-dependent materials. Experiments show that, compared to prior works handling reflective surfaces, our method strikes a commendable balance between efficiency and visual quality, avoiding the massive memory footprints that plague existing specular-aware Gaussian methods.

In summary, our proposed method offers several significant advantages:

- 1) Our method explicitly decouples scene appearance into diffuse and specular components using a Hybrid Representation (SH+ASG), significantly enhancing the realism of reflective surfaces.
- 2) We propose a novel Reflection-Prior Guided Densification (Ref_Score) and Role Separation Supervision framework that intelligently allocates computational resources, ensuring stable and precise optimization.
- 3) Leveraging the efficiency of the FastGS backbone, our approach completely resolves the trade-off between specular reconstruction quality and computational cost, providing true real-time rendering capabilities for interactive applications.

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